ECO 3121 2024 – 2025 Fall Assignment 2

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ECO3121: Introductory Econometrics

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Assignment 2

Foreword:

This is the Answer to Assignment 2, written by Junpeng Liang (Student ID: 123090319), combining all the required answers, Stata code, and requested output. Do.file is attached in the last.

Stata Codes and **Required Output** will be shown in the very front of each Question.

Short Answers will be shown after Required Output one by one to Answer the Related Questions.

Question 1

1. Run the regression and report the estimated coefficients and their standard errors, and comment on the statistical significance of the regression estimates and R^2.

Stata Codes:

reg wage yeduc nin

Required Output:

Intercept (_cons): <u>72117.21</u> (Std. Err.: 2519.881)

Years of Education (yeduc): <u>17826.42</u> (Std. Err.: 193.9603)

Nin (nin): <u>-6880.068</u> (Std. Err.: 714.7795)

```
. reg wage yeduc nin
```

Source	SS	df	MS	Number of obs	=	61,136
				F(2, 61133)	=	4318.10
Model	3.2226e+14	2	1.6113e+14	Prob > F	=	0.0000
Residual	2.2812e+15	61,133	3.7315e+10	R-squared	=	0.1238
				Adj R-squared	=	0.1238
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	1.9e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	17826.42	193.9603	91.91	0.000	17446.26	18206.59
nin	-6880.068	714.7795	-9.63	0.000	-8281.038	-5479.099
_cons	72117.21	2519.881	28.62	0.000	67178.24	77056.19

Statistical Significance:

For variable yeduc: t-value = 91.91, P>|t| = 0.000.

This indicates that the effect of education on wage is statistically significant at any common significance level (α) equals to 0.01 or 0.05 or 0.10.

For variable nin: t-value = -9.63, P>|t| = 0.000.

This indicates that the effect of nin on wage is statistically significant at any conventional significance level (α) equals to 0.01 or 0.05 or 0.10.

Interpretation of Coefficients:

Predictably, an increase of one unit in education is associated with an increase in wage of approximately 17826.42 units under the ceteris paribus (holding variable nin constant). Showing that independent variable yeduc has positive impact on wage.

Predictably, an increase of one unit in nin is associated with a decrease in wage of approximately 6880.068 units under the ceteris paribus (holding variable yeduc constant). Showing that independent variable nin has negative impact on wage.

Overall Model Significance:

The F-statistic (4318.10) with a p-value of 0.000 showcases that the overall model is statistically significant, which means that at least one predictor variable significantly predicts wages for this model.

Comment On R^2:

R^2 = 0.1238.

R^2 indicates that the model explains approximately 12.38% of the independent variables in the dependent variable wage. While both predictors are statistically significant and the model is significant, the low R^2 value means that some other factors except variables yeduc and nin may also affect wage.

2. Comparing the estimates from these two models (specifications), What would be the bias, and in which direction of the misspecified model? Be as precise as possible to explain the comparison.

Stata Codes:

reg wage yeduc

Required Output:

Intercept (_cons): 57405.06 (Std. Err.: 2004.897)

Education (yeduc): 17910.03 (Std. Err.: 193.9109)

```
. reg wage yeduc
```

Source	SS	df	MS	Number of obs	=	61,136
				F(1, 61134)	=	8530.77
Model	3.1880e+14	1	3.1880e+14	Prob > F	=	0.0000
Residual	2.2846e+15	61,134	3.7371e+10	R-squared	=	0.1225
				Adj R-squared	=	0.1224
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	1.9e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	17910.03	193.9109	92.36	0.000	17529.96	18290.09
_cons	57405.06	2004.897	28.63	0.000	53475.45	61334.66

Direction of Bias:

Coefficient of yeduc would be biased.

In Model 1, when nin is included, the coefficient for education is 17826.42.

In Model 2, when nin is excluded, the coefficient for education is 17910.03, which increases by 83.61.

The increase shows the direction is upward.

Explanation of the Bias:

Since we can know from the Model 1 that variable nin has negative effect of on wage (-6880.068). When variable nin with a negative coefficient is omitted, the coefficient of yeduc is overestimated. This is usually due to “omitted variable bias”. Variable nin has a negative effect on wage, but this negative effect is not accounted for in the error model, so the model tries to compensate for the error due to the omission of nin by assigning higher weights to yeduc.

Conclusion:

The coefficients from Model 2 are biased due to the omission of nin, resulting in an upward bias for the coefficient of yeduc.

3. Based on the estimates and comparison from these two models, what's the direction of cov(education ninja) from theoretical judgment? Based on the dataset itself, what is the direction of cov($education_i$ nin_j)? Are these two results consistent with each other?

Stata Codes:

corr yeduc nin

cov ($education_i$, nin_j) = -0.0621

```
. corr yeduc nin
(obs=152,989)
```

	yeduc	nin
yeduc	1.0000	
nin	-0.0621	1.0000

Based on the estimates and comparison from these two models:

Since we know if we exclude the variable nin, the coefficient of yeduc will increase. We can deduce that corr($education_i$, nin_j) are negatively correlated, and the cov($education_i$, nin_j) should be negative.

Based on Dataset:

Using the STATA, we can easily find out the corr(education$_i$, nin$_j$) is negative, meaning that variable yeduc and variable nin are negatively correlated, and the covariance should be negative(since the variance of both variable yeduc and nin is larger than 0). This suggests that as the number of inpres schools increases, the average level of education decreases, or vice versa. These two results are consistent with each other.

4. A professor thinks about adding an additional variable en71 to model 1, and asks that does the enrollment rate in 1971 of each province also change the wage outcome? Run the regressions recommended by the professor. Write down the new model (specification) as model 3.

Stata Codes:

reg wage yeduc nin en71

Required Output:

$$\mathbf{wage_{ij} = \alpha + \beta_1\, education_i + \beta_2\, nin_j + \beta_3\, en71 + \mu_i}$$

```
. reg wage yeduc nin en71
```

Source	SS	df	MS	Number of obs	=	60,936
				F(3, 60932)	=	2873.13
Model	3.2216e+14	3	1.0739e+14	Prob > F	=	0.0000
Residual	2.2774e+15	60,932	3.7376e+10	R-squared	=	0.1239
				Adj R-squared	=	0.1239
Total	2.5996e+15	60,935	4.2661e+10	Root MSE	=	1.9e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	17704.64	197.781	89.52	0.000	17316.98	18092.29
nin	-7513.557	735.1522	-10.22	0.000	-8954.457	-6072.656
en71	35557.16	9684.733	3.67	0.000	16575.06	54539.27
_cons	68233.96	2736.359	24.94	0.000	62870.69	73597.23

5. Formally write down the corresponding hypothesis testing (t-test) to verify the professor's question step by step (5 steps). And please make your conclusion to the professor's question.

Traditional approach to hypothesis testing:

- Choose a null hypothesis: $H_0 : \beta_j = 0$
- Choose an alternative hypothesis: $H_1 : \beta_j > 0$
- Choose a **significance level** (or simply **level**, or **size**) for the test. That is, the probability of rejecting the null hypothesis when it is in fact true (**Type I Error**) .
- Calculate t-statistics $t_{\hat{\beta}_j}$
- Choose a critical value, $c > 0$, so that the **rejection rule** is

$$t_{\hat{\beta}_j} > c$$

Hypothesis Test Steps:

Step 1: Given the null hypothesis: H_0: $\beta_1 = 0$, $\beta_2 = 0$, $\beta_3 = 0$

Step 2: Given the alternative hypothesis H_1: $\beta_1 \neq 0$, $\beta_2 \neq 0$, $\beta_3 \neq 0$

Step 3: Choose a significance level, such as $\alpha = 0.05$.

Step 4: Calculate t-statistics: t = en71_hat / se(en71_hat)

Step 5: Compare it against the critical t-value from the t-distribution for the relevant degrees of freedom.

Conclusion:

From the previous Models, we can find out P-values of yeduc, nin and en71 are all 0.000, and are all less than 0.01, 0.05, 0.10.

Thus, we can reject H_0 at a significance level equal to 0.01, 0.05, 0.10.

We can conclude: The enrollment rate in 1971 of each province indeed also changed the wage outcome at some degree.

6. Comparing model 3 and model 2, the professor wants to test the joint significance of variables nin and en71. What kind of test shall we use? Please offer the detailed procedure for the test step by step and conclude.

Stata Codes:

test nin en71

Required Output:

```
. test nin en71

 ( 1)  nin = 0
 ( 2)  en71 = 0

       F(  2, 60932) =   53.17
            Prob > F =    0.0000
```

Using the F-statistic:

Since we want to find out the joint significance o. 2 variables nin and en71.

The detailed procedure for the test:

By using the Stata. We can find out the F-statistic is 53.17.

The degrees of freedom for the numerator are 2 (df_r - df_{ur}).

The degrees of freedom for the denominator are 60932 (df_{ur}).

The p-value related with this F-statistic is 0.0000.

To determine whether to reject the null hypothesis or not, we compare the p-value to the significance level (Here we choose $\alpha = 0.05$):

$$\text{P–value} = 0.000 < \alpha = 0.05$$

Reject the Null Hypothesis (H_0) at a significance level $\alpha = 0.05$

Conclusion:

Since the p-value is 0.0000, which is significantly less than 0.01, 0.05, 0.10. We can reject the null hypothesis at significance level equals to 0.01, 0.05, 0.10.

This shows that at least one of the coefficients for nin and en71 is significantly different from zero, suggesting that the variables contribute strongly to the model.

Question 2

1. Please run a regression of Y: high wage on X_i: years of education (yeduc) and Z_j: number of inpres school per children (nin) using 1) linear probability model and 2) Probit model, respectively.

   Summarize these estimates in a table with 3 columns (the format of the table can refer to the table on page 24 of the week 8-2 slide, but the information in the last row of that table does not need to be displayed); and only interpret the coefficient of yeduc in the linear probability model.

Stata Codes:

gen high_wage = (wage >= 192000 & wage < .)

reg high_wage yeduc nin

probit high_wage yeduc nin

Required Output:

	Linear	**Probit**
yeduc	**.0359757**	**.1431199**
nin	**-.0133587**	**-.0528456**
_cons	**-.0601791**	**-2.020965**

```
. reg high_wage yeduc nin
```

Source	SS	df	MS	Number of obs	=	152,989
				F(2, 152986)	=	12581.40
Model	3461.70477	2	1730.85239	Prob > F	=	0.0000
Residual	21046.6388	152,986	.137572319	R-squared	=	0.1412
				Adj R-squared	=	0.1412
Total	24508.3435	152,988	.160197816	Root MSE	=	.37091

high_wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
yeduc	.0359757	.0002298	156.52	0.000	.0355252	.0364261
nin	-.0133587	.000836	-15.98	0.000	-.0149972	-.0117202
_cons	-.0601791	.0027973	-21.51	0.000	-.0656618	-.0546964

```
. probit high_wage yeduc nin

Iteration 0:  Log likelihood = -76625.607
Iteration 1:  Log likelihood = -65446.635
Iteration 2:  Log likelihood = -65267.143
Iteration 3:  Log likelihood = -65267.009
Iteration 4:  Log likelihood = -65267.009

Probit regression                                 Number of obs =  152,989
                                                  LR chi2(2)    = 22717.20
                                                  Prob > chi2   =   0.0000
Log likelihood = -65267.009                       Pseudo R2     =   0.1482
```

high_wage	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
yeduc	.1431199	.0010296	139.01	0.000	.1411019	.1451379
nin	-.0528456	.0035573	-14.86	0.000	-.0598179	-.0458734
_cons	-2.020965	.0131103	-154.15	0.000	-2.046661	-1.995269

Interpretation of the coefficient of yeduc in the linear probability model:

The coefficient of yeduc in the linear probability model is 0.035978, which means that for each additional year (units) of education, the probability of receiving a high wage will increase by approximately 3.60% under the ceteris paribus (holding variable nin constant).

This positive relationship shows that higher education is associated with a greater probability of earning a high wage.

2. The professor uses the following Probit model. Discuss 2-3 potential drawbacks of using a linear probability model instead of a non-linear model when Y is a binary variable.

Drawbacks:

1. **LPM may predict probabilities outside the [0,1] range, which is not meaningful for binary outcomes and is not suited to reality.**
2. **LPM has constant error variance (Assumptions in OLS), which usually fails in binary outcomes, leading to inefficiency.**
3. **The relationship between explanatory variables and the dependent variable may be non-linear when LPM only focuses on a linear model and easily fails to fit.**

3. Use the Probit model and the results from the table you offer in question (1) and calculate the change in probability when Z = 3 and X increases from 8 to 12. (Note: you can keep () in your answers without solving for it.)

Stata Codes:

```
scalar p1 = normal(-2.020965 + .1431199*12 + -.0528456*3)

scalar p2 = normal(-2.020965 + .1431199*8 + -.0528456*3)

scalar change = p1 - p2

display change
```

Required Output:

1.8495587

```
. scalar p1 = normal(-2.020965 + .1431199*12 + -.0528456*3)

. scalar p2 = normal(-2.020965 + .1431199*8 + -.0528456*3)

. scalar change = p1 - p2

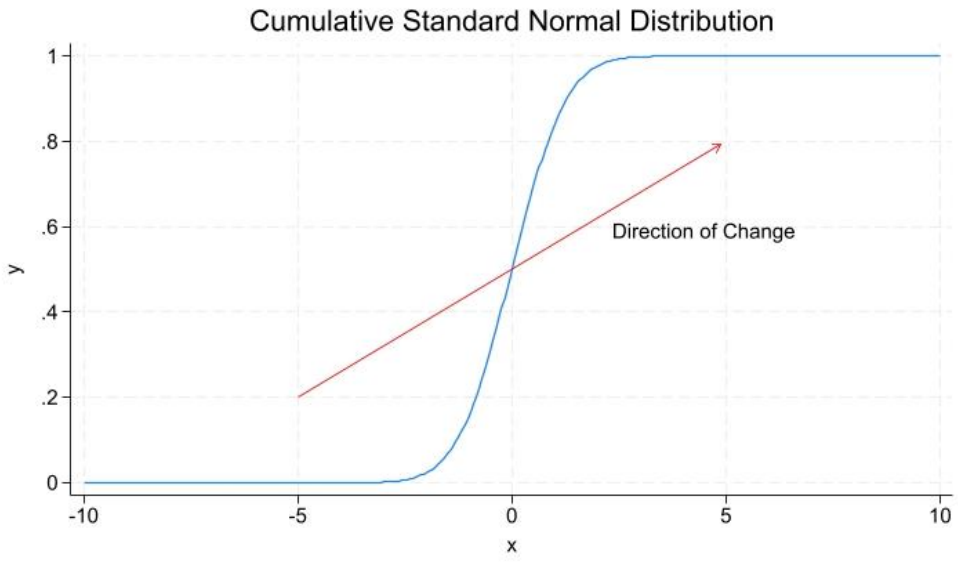
. display change
1.8495587
```

4. Use a graph of the cumulative standard normal distribution to demonstrate such change in probability in question (3) and label the direction of the change. (Note: no need to spend time drawing a very neat figure. The information conveyed by this figure is more important.)

Stata Codes:

twoway function y = normal(x), range(-10 10) title("Cumulative Standard Normal Distribution")

Required Output:



Question 3

1. Run a regression of monthly wages on high level. Demonstrate the coefficient of high level and interpret the estimate.

Stata Codes:

gen high_level = (yeduc >= 9 & yeduc < .)

reg wage high_level

Required Output:

High_level: 112764.3 (Std. Err.: 1669.638)

```
. reg wage high_level
```

Source	SS	df	MS	Number of obs	=	61,136
				F(1, 61134)	=	4561.41
Model	1.8076e+14	1	1.8076e+14	Prob > F	=	0.0000
Residual	2.4227e+15	61,134	3.9629e+10	R-squared	=	0.0694
				Adj R-squared	=	0.0694
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	2.0e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
high_level	112764.3	1669.638	67.54	0.000	109491.8	116036.8
_cons	156631.9	1327.543	117.99	0.000	154029.9	159233.9

Interpretation:

The coefficient of high level is 112764.3. The coefficient for high_level shows the average difference in monthly wages between those with 9 or more years of education and those with less than 9 years of education.

The positive coefficient suggests that higher education (Particularly those with more than 9 years) is associated with higher wages.

2. The professor wants to study whether urban as the birthplace (urban = 1 in the data) can affect the slope of a high level of wage by introducing the interaction term.

   1) Please write down the model (specification) and report your estimates. (Don't forget to include urban as an independent variable in your model.)

   2) Conduct your hypothesis testing using p-value, and conclude.

$$\hat{wage} = \beta_0 + \beta_1 * highlevel + \beta_2 * urban + \beta_3 * (highlevel * urban)$$

Stata Codes:

gen interaction = high_level * urban

reg wage high_level urban interaction

Required Output:

wage = 152485.9 + 99802.76 * high_level + 25014.22 * urban + 23411.8 * (high_level * urban)

```
. reg wage high_level urban interaction
```

Source	SS	df	MS	Number of obs	=	61,136
				F(3, 61132)	=	1727.11
Model	2.0342e+14	3	6.7805e+13	Prob > F	=	0.0000
Residual	2.4000e+15	61,132	3.9259e+10	R-squared	=	0.0781
				Adj R-squared	=	0.0781
Total	2.6034e+15	61,135	4.2585e+10	Root MSE	=	2.0e+05

wage	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
high_level	99802.76	1914.016	52.14	0.000	96051.29	103554.2
urban	25014.22	3553.394	7.04	0.000	18049.56	31978.88
interaction	23411.8	4131.893	5.67	0.000	15313.28	31510.32
_cons	152485.9	1446.661	105.41	0.000	149650.4	155321.3

Hypothesis Test Steps:

Step 1: Given the Null Hypothesis: The coefficient of interaction is equal to 0 (has no effect on the wage).

$$H_0: \beta_3 = 0$$

Step 2: Given the Alternative Hypothesis: The coefficient of interaction is not equal to 0 (has an effect on the wage).

$$H_1: \beta_3 \neq 0$$

Step 3: Choose a significance level. Here we choose most common significance level.

$$\alpha = 0.05$$

Step 4: Calculate t-statistics:

P-value = 0.000

Step 5: Compare it against the critical t-value from the t-distribution for the relevant degrees of freedom.

$$\text{P-value} = 0.000 < \alpha = 0.05$$

Conclusion:

P-value (0.000) is less than the significance level (0.05).

Thus, we can reject the Null hypothesis(H_0) at the significance level $\alpha = 0.05$.

Showing that the slope of high_level on wage is affected by being urban.